

Playfair Cipher – PT-109 Message

9. Write a C program for PT-109 American patrol boat, under the command of Lieutenant John F. Kennedy, was sunk by a Japanese destroyer, a message was received at an Australian wireless station in

Playfair code:

KXJEY UREBE ZWEHE WRYTU HEYFS

KREHE GOYFI WTTTU OLKSY CAJPO

BOTEI ZONTX BYBNT GONEY CUZWR

GDSON SXBOU YWRHE BAAHY USEDQ

Aim

To write a C program to decrypt the given **Playfair cipher** using the key:

ROYAL NEW ZEALAND NAVY

The standard Playfair square for this key is: [FAUbox](#)

R O Y A L

N E W Z D

V B C F G

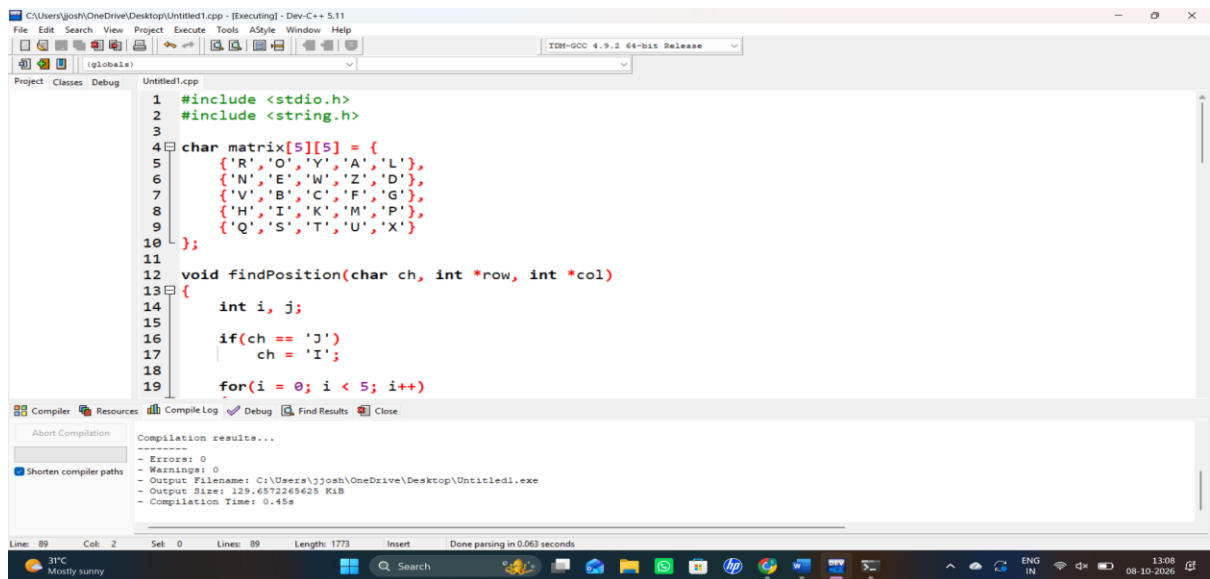
H I K M P

Q S T U X

Algorithm

1. Construct the 5×5 Playfair matrix.
2. Combine I/J.
3. Divide ciphertext into pairs.
4. If both letters are in the same row, move left.
5. If both are in the same column, move up.
6. Otherwise, form a rectangle and exchange columns.
7. Display plaintext.

Program

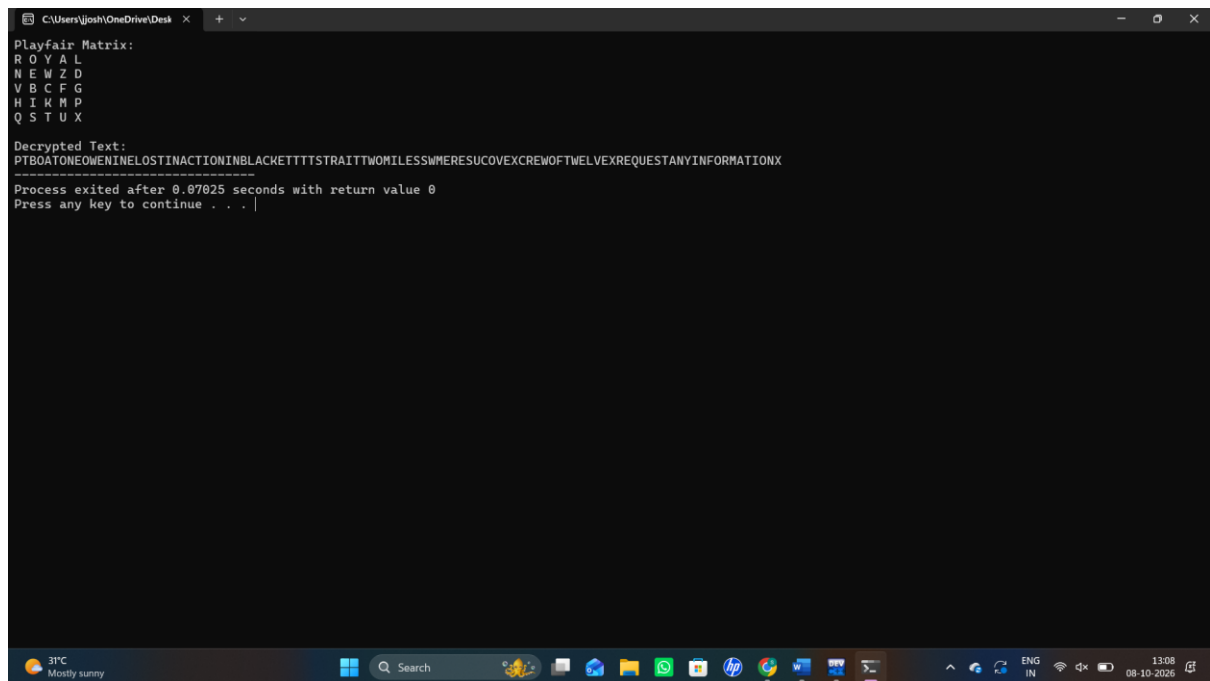


```
1 #include <stdio.h>
2 #include <string.h>
3
4 char matrix[5][5] = {
5     {'R','O','Y','A','L'},
6     {'N','E','W','Z','D'},
7     {'V','B','C','F','G'},
8     {'H','I','K','M','P'},
9     {'Q','S','T','U','X'}
10 };
11
12 void findPosition(char ch, int *row, int *col)
13 {
14     int i, j;
15
16     if(ch == 'J')
17         ch = 'I';
18
19     for(i = 0; i < 5; i++)
```

Compilation results...

- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\josh\OneDrive\Desktop\Untitled1.exe
- Output Size: 129.6572265625 KiB
- Compilation Time: 0.45s

Input/Output



```
Playfair Matrix:
R O Y A L
N E W Z D
V B C F G
H I K M P
Q S T U X

Decrypted Text:
PTBOATONEOWENINELOSTINACTININBLACKETTTTSTRAITTWOMILESSMMERESUCOVEXCREWOFTWELVEXREQUESTANYINFORMATIONX

Process exited after 0.07025 seconds with return value 0
Press any key to continue . . .
```

Result

Thus, the given PT-109 ciphertext was successfully decrypted using the Playfair cipher.